

How to Tackle System Development Life Cycle (SDLC) Using Techniques in Analysis and Design Phase

Farra Nurzahin binti Zaharil Anuar, Harini A/P Sangaran

Abstract—This study explores the use of techniques from the system analysis and design phases to successfully tackle the System Development Life Cycle (SDLC). Beginning with an explanation of key system analysis techniques such as problem identification and Cost-Benefit Analysis (CBA) and emphasises the necessity of detecting inefficiencies and determining economic feasibility. Furthermore, this paper looks into design phases including prototyping, interface design and database design which emphasise their importance in turning system specifications into functional systems. By using these techniques, organisations can improve system efficacy, economic viability and user experience throughout the SDLC process. This paper aims to provide useful insights and practical recommendations to professionals and scholars looking to optimise the SDLC utilising methodologies from both system analysis and design phases.

Keywords—Techniques, System Development Life Cycle, SDLC, System Analysis and Design, Analysis Phases, Design Phases.

I. INTRODUCTION

System Development Life Cycle (SDLC) is a conceptual model used in project management that describes the stages involved in an information system development project [7]. Traditionally, the SDLC consists of seven phases, which are problem identification, determining user requirements, analysing system needs, designing the system, developing and documenting software, testing and maintaining system and lastly, implementing and evaluating system. Each of this phases plays a crucial role in producing an efficient and effective final product according to the users needs and wants. In our article today, we'll be discussing about the techniques used by system analysts in analysis and design phase of SDLC.

II. TECHNIQUES IN ANALYSIS AND DESIGN PHASE

A. Techniques in System Analysis

System analysis phase is a crucial phase in SDLC. System analysis in SDLC includes a draft proposal for a system that contains a summary of the users, functionality and value of the current systems, a cost-benefit analysis of the alternatives and recommendations for what should be done [1]. In this critical phase, various techniques are employed to facilitate effective design and development.

One of the techniques in the system analysis phase is identifying problems. This technique entails carrying out comprehensive analyses to identify current problems, defects or inadequacies in the systems or procedures in place. Identifying the current problem involves sorting through the majority of the problems in order to locate as many issues as possible and determine the issue that led to the majority of

these problems [2]. Through identifying and recording these issues, analysts can get important knowledge regarding the underlying reasons for inefficiencies and areas of improvement. Besides, [3] state that inadequate problem analysis can also lead to disadvantages of the solutions model. This technique can be effectively employed by utilizing the 3W2H questions by asking questions like 'What are the issues preventing the completion of activities?' and 'How do these problems contribute to the adequacy of activities?' [2]. Through the use of the 3W2H framework, challenges can be thoroughly and systematically analysed, leading analysts to practical conclusions for system enhancement within the SDLC.

In addition, another technique required in system analysis is by using a Cost-Benefit Analysis (CBA) technique. This technique involves determining the system's possible implementation costs and benefits in order to evaluate its economic feasibility [4]. The necessity to evaluate the possible return on investment and justify investments grows as software systems get more complicated and resource-intensive. The five parts of this technique procedure are often some form of identifying the project scope, calculating costs, calculating benefits, computing analysis calculations and formulating and implementing recommendations [4]. These processes determine if the benefits of a system outweigh the system's cost limitations before deciding whether to launch or consider alternatives options. These claims are strongly agreed by [5] that CBA is a very powerful tool for system analysis because it adds into an extensive sensitivity analysis and full justification of all assumptions and estimations. Additionally, cost-benefit analysis helps with risk evaluation and reduction plans, which helps businesses set priorities for costs and distribute resources wisely.

Moreover, in the system analysis stage of the SDLC, pinpointing areas where the current system can be improved and developing specifications for a new system are essential techniques. This technique usually makes system specifications to convert into a working system throughout the installation, testing and coding phases of the system's implementation and operation, guaranteeing its preparedness for use [6]. During the coding phase, programmers create the programmes that comprise the system, while in the testing phase, programmers and analysts look over individual programmes as well as the system as a whole to find and fix bugs. This testing phase not only guarantees the system's performance and dependability but also gives end users confidence in its use and functionality. Lastly, at the installation phase, the new technology is integrated into the organization's regular operations [6]. Furthermore, it should be noted that system implementation can be continued indefinitely because continuing user assistance is also included.

B. Techniques Used in Design Phase

Just like the analysis phase of SDLC, design phase also has its techniques to be applied so that an effective and efficient outcome is produced. One of the techniques used to ease the process in design phase is called prototyping. Prototyping is known as designing and building a scaled-down but working version of a desired system [1]. Prototypes are created during the analysis phase of SDLC so that the analysts can study and understand the needs and wants of the users of their product. well, how does it relate to design phase then? The final product that is created in the design phase is the improvement and completion of the prototype. Prototypes are just mere demonstration of the product that is going to be produced which might lack a few touches of the final product. However, design phase is not as easy as it seems. There are multiple factors to be considered to produce a functional design.

Design phase can be segregated into designing interfaces and designing databases. Let's look at designing interfaces first. The purpose of an interface is to deliver and capture information from users [6]. An effective human-computer interface allows the users to find, view and invoke the different components of a system. To design a good interface, there are multiple steps to follow.

Firstly, designing layouts to record data from users for your interface. Your computer-based forms must be similar to paper-based forms and the flow of the form must be from left-to-right and from top-to-bottom to ease the process of filling up data. Figure 1 shows an example of a good computer-based form layout [6].

Customer Order Report Today: 11-OCT-12
Order Number: 913-A36-98

Customer Information

Customer Number:	1273
Name:	Contemporary Designs
Address:	123 Oak Street
City:	Austin
State:	TX
Zip:	28384

PRODUCT NUMBER	DESCRIPTION	QUANTITY ORDERED	UNIT PRICE	TOTAL PRICE
M128	Bookcase	4	200.00	800.00
B381	Cabinet	2	150.00	300.00
B210	Table	1	500.00	500.00
G200	Deluxe Chair	8	400.00	3,200.00

TOTAL ORDER AMOUNT	4,800.00
5% DISCOUNT	240.00
TOTAL AMOUNT DUE	4560.00

Help Print (password required) Select Customer or Exit

Figure 1

Next, a system analyst should control the data input when allowing the users to provide data. This simply means that system analysts should anticipate and tackle several types of errors made by users such as adding extra data on a field or providing wrong characters in a field. For example, users are required to key-in their phone number whereas a user enters an alphabet in the column which results in invalid data.

Last but not least, interfaces should also provide feedback in order to facilitate the use of it for users. System feedback can be divided into three types which are status information, prompting cues and showing error or warning messages [1]. Status information such as "Loading..." or "The processing is taking longer than expected, please be patient." not only allows users to know what is the system doing but also feel engaged with the system. Moreover, prompting cues such as "Please enter your phone number :'" shows that the system is ready for input. You can also improve the design by requesting for specific format such as "Please enter your phone number (01X-XXXXXXX) :'" which shows how you want your data to be entered. Showing error or warning message on the other hand allows users to recognize their mistakes and correct it.

As mentioned earlier, design phase also consists of designing databases. The purpose of creating a database is for it to store, retrieve and manage the data [6]. SQL Server is one of the examples of a database design. Firstly, the system analyst needs to decide on the data that needs to be stored. Then, create a table and define the structure of each table. For example, employee ID as an attribute that you might want to define. Next, the data of the users should be filled under each attribute. Once you have filled up the data in the table, you will have to create relationships between multiple tables where an attribute in Table 1 is also related in Table 2. Then, normalization takes place. Normalization is the conversion of the complex data structures into simple and stable data structures [6]. Normalization is also a process that eliminates repeating groups found in the records [1]. Based on the definition, it is clear that data is normalized to reduce the redundancy of the data so that the usage is efficient. Last but not least, a system analyst should add constraints to their database where certain data have to follow certain format to ensure data efficiency and integrity. Design phase can be segregated into designing interfaces and designing databases.

III. CONCLUSION

To conclude, it is evident that multiple thought processes and techniques are considered in both analysis and design phase. Design phase includes designing interfaces and designing databases. Designing interfaces starts with designing a layout, then controlling data input and finally providing feedback whereas designing database is about storing and managing data. These techniques are used solely to improve the quality of work and also to enable system analysts to have a proper guideline. Therefore, system analysts are recommended to follow all the phases of SDLC and use these techniques to tackle SDLC better.

REFERENCES

- [1] K. E. Kendall and J. E. Kendall, "System Analysis and Design", 8th ed. Inc., Pearson Education, 2011. [E-book] Available : <https://www.auhd.edu.ye/upfiles/elibrary/Azal2020-01-22-12-35-12-90529.pdf> [Accessed March 25, 2024]
- [2] O. Onasuk, P. Wuttidittachotti, S. Prakanchaoen, S. A. O. Vallipakorn, "A SDLC developed software testing process using DMAIC model", ARPN J. Eng. Appl. Sci, 10(3), p. 1008-1016, February, 2015. [Online]. Available: https://www.researchgate.net/profile/Sakda-Vallibhakara/publication/281107879_A_SDLC_developed_software_testing_process_using_DMAIC_model/links/55d6e09708aec156b9a882c4/A-SDLC-developed-software-testing-process-using-DMAIC-model.pdf . [Accessed April 3, 2024]
- [3] R. Arora & N. Arora "Analysis of SDLC models", International Journal of Current Engineering and Technology, 6(1), p. 268-272, February 2016. [Online]. Available: <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=8>

97f5e3f37adfc96fb09d1f72cf81cf65331f588 . [Accessed April 3, 2024]

- [4] A. Hayes ,”What Is Cost-Benefit Analysis? How Is It Used? What Are its Pros and Cons?”, February, 2024. [Online]. Available: <http://www.investopedia.com/terms/c/cost-benefitanalysis.asp> . [Accessed at March 29, 2024]
- [5] S. R. Cellini, J. E. Kee, (2015). “Cost - effectiveness and cost - benefit analysis ” , p. 636-672, 2015. [Online]. Available: [https://d1wqtxts1xzle7.cloudfront.net/38126636/CelliniKee21-libre.pdf?1436340742=&response-content-disposition=inline%3B+filename%3Dw_CHAPTER_TWENTY_ON_E_COST_EFFECTIVENESS.pdf&Expires=1711948855&Signature=bXiNWCCUaO~IXV9DN6-DLCCNyJA5rUQ~FWIdZZHtg1JkN0yC8tOg0RqdnvJM6wBD9n~LkrSOO2CrdNdBQq3FaCqUUGgTFF0R74Ny1AdXkOh1luUQGvKl2RCsnqsTl7bHVUiMsdmIFFtC7sCk1uggrj3zEPprCJJC~iWjDy2TDy](https://d1wqtxts1xzle7.cloudfront.net/38126636/CelliniKee21-libre.pdf?1436340742=&response-content-disposition=inline%3B+filename%3Dw_CHAPTER_TWENTY_ON_E_COST_EFFECTIVENESS.pdf&Expires=1711948855&Signature=bXiNWCCUaO~IXV9DN6-DLCCNyJA5rUQ~FWIdZZHtg1JkN0yC8tOg0RqdnvJM6wBD9n~LkrSOO2CrdNdBQq3FaCqUUGgTFF0R74Ny1AdXkOh1luUQGvKl2RCsnqsTl7bHVUiMsdmIFFtC7sCk1uggrj3zEPprCJJC~iWjDy2TDyH2FwacXa1z~6ZiBeZHZxAAMC48tt4O-AkluEVy8xQk7sk5B9RcQR0mMzzXAEepwqbzPaBylOQU7Bd0ziC0-WUxEouYigszST-ju3~yNNYskUgI7nxFQ7TF--qvjcHx5NWFN1KjATPHsdHrC-DsVs3cPCQaLoAc-nFZaAAhAYfqTg__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA)
- [6] J. S. Valacich, J. F. George and J. A. Hoffer, "Essentials of System Analysis and Design", 5th ed. Inc., Pearson Education, 2012. [E-book] Available : https://www.awa2el.net/sites/default/files/essentials_of_systems_analysis_and_design_5th_by_valacich_.pdf [Accessed March 25, 2024]
- [7] A. S. Gillis, “Systems Development Life Cycle (SDLC)”, June 2019. [Online]. Available : <https://www.techtarget.com/searchsoftwarequality/definition/systems-development-life-cycle> [Accessed April 3, 2024].